WORKSHEET 4.1

What a Derivative Means, in Words and Units

Unit 4: Derivatives in Context

Key ideas

- $f'(a)$ is the **rate** at which f is changing at the instant $x = a$. Its units are the units of f per unit of x.
- A complete interpretation names the instant, the quantity, whether it is increasing or decreasing, the rate (as a positive number) and the units.
- If f is itself a rate, f' is a rate of change of a rate: for example litres per minute *per minute*.
- A value and a rate give a nearby estimate: $f(a+h) \approx f(a) + f'(a)\,h$.

Worked examples

Example 1 · Interpreting

The cost of producing q kilograms of flour is $C(q)$ dollars. Interpret $C'(40) = 3.5$.

Solution. The units are dollars per kilogram, at the production level $q = 40$.

When 40 kg are being produced, the cost is increasing at 3.5 dollars per additional kilogram.

Example 2 · Estimating

A plant has height $h(t)$ cm after t weeks, with $h(6) = 30$ and $h'(6) = 2.5$. Estimate its height at 8 weeks.

Solution. $h(8) \approx 30 + 2.5(2)$.

About 35 cm.

Example 3 · A rate of a rate

A car's velocity is $v(t)$ metres per second at time t seconds. State the units of $v'(3)$ and interpret $v'(3) = -1.2$.

Solution. Metres per second per second.

At $t = 3$ seconds the velocity is decreasing at 1.2 metres per second each second.

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Practice

CORE

1. The volume of water in a tank is V litres at time t hours. State the units of $\dfrac{dV}{dt}$.

2. The temperature of a drink is $T(t)$ degrees Celsius after t minutes. Interpret $T'(15) = -0.8$.

3. Given $A(20) = 150$ and $A'(20) = 4$, estimate $A(22)$.

4. What does $f'(5) < 0$ tell you about f at $x = 5$?

EXAM LEVEL

5. The revenue from selling x items is $R(x)$ dollars. Interpret $R'(100) = 1.5$.

6. A car has $F(t)$ litres of fuel in its tank after t hours of driving. Interpret $F'(2) = -7$.

7. Gas flows into a cylinder at the rate $G(t)$ litres per minute. State the units of $G'(4)$ and interpret $G'(4) = 0.3$.

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8. A patient's weight is $W(t)$ kg after t weeks, with $W(10) = 65$, $W'(10) = -0.5$ and $W''(10) = 0.02$. Estimate $W(12)$, and say whether the weight loss is speeding up or slowing down at $t = 10$.

Question 9 · Free-response practice

The depth of snow on a field is $S(t)$ centimetres, where t is the number of hours after midnight and S is differentiable with a continuous derivative. It is known that $S(3) = 12$, $S'(3) = 1.8$ and $S'(9) = -0.6$.

(a) Using correct units, interpret $S'(3) = 1.8$.

(b) Use a tangent line to estimate the depth of snow at $t = 4$.

(c) Explain why there must be a time between $t = 3$ and $t = 9$ at which the depth of snow is neither increasing nor decreasing.

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Worked solutions

1. Litres per hour.
2. At $t = 15$ minutes the temperature of the drink is decreasing at 0.8 degrees Celsius per minute.
3. $150 + 4(2) = 158$.
4. f is decreasing at $x = 5$.
5. When 100 items have been sold, revenue is increasing at 1.5 dollars per additional item.
6. After 2 hours of driving, the amount of fuel in the tank is decreasing at 7 litres per hour.
7. Litres per minute per minute. At $t = 4$ minutes the rate at which gas flows in is increasing by 0.3 litres per minute each minute.
8. $W(12) \approx 65 - 0.5(2) = 64$ kg. The rate W' is negative and increasing (becoming less negative), so the weight loss is slowing down.
9. (a) At 3 a.m. the depth of snow is increasing at 1.8 centimetres per hour.
   (b) $S(4) \approx 12 + 1.8(1) = 13.8$ cm.
   (c) S' is continuous with $S'(3) > 0$ and $S'(9) < 0$, so by the Intermediate Value Theorem $S'(c) = 0$ for some c between 3 and 9.

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